

5. RSA ALGORITHM

AIM

To implement the RSA public key cryptography algorithm using Python.

Algorithm

1. Start the program.
2. Generate RSA public and private keys.
3. Enter the plaintext.
4. Encrypt the plaintext using the public key.
5. Display the ciphertext.
6. Decrypt the ciphertext using the private key.
7. Display the original plaintext.
8. Stop.

Python Code

```
from Crypto.PublicKey import RSA
from Crypto.Cipher import PKCS1_OAEP
import base64

message = input("Enter the Plain Text: ")
key = RSA.generate(2048)
public_key = key.publickey()
private_key = key
encryptor = PKCS1_OAEP.new(public_key)
ciphertext = encryptor.encrypt(message.encode())
print("\nEncrypted Text:")
print(base64.b64encode(ciphertext).decode())
decryptor = PKCS1_OAEP.new(private_key)
plaintext = decryptor.decrypt(ciphertext)
print("\nDecrypted Text:")
print(plaintext.decode())
```

Output

```
*** Enter the Plain Text: i am jake

Encrypted Text:
VmYgJfnLjPvTox4Ybj1jThRjX8pqkahJDMZUCiC1bb2mEar8vVJ57jcU3DV/zLRALxFLVnOD6nDIriGdFRuf2QR3/wtGZWS92UUMKc1W81+YARFUNFviiH3Mk1JR6KIuR/tamOa2xzk

Decrypted Text:
i am jake
```

The result:

The above code is executed and the output is obtained successfully.